

Fedja Bogataj

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Education

- **Technical University of Vienna (TU Wien)** — Vienna, Austria
  Master of Science in Data Science; GPA: 1.23 — *Mar. 2025 – Present*
  - **Specialization**: Machine Learning and Statistics, Big Data and High Performance Computing
  - **Relevant Coursework**: Machine Learning, Advanced Methods for Regression and Classification, Statistical Simulation and Computer-intensive Methods, AKSTA Statistical Computing, Experiment Design and Execution
- **University of Ljubljana, Faculty of Computer and Information Science** — Ljubljana, Slovenia
  Bachelor of Computer Science; Avg. grade: 9.00/10 — *Oct. 2021 – Sep. 2024*
  - **Specialization**: Machine Learning and Artificial Intelligence
  - **Relevant Coursework**: Probability and Statistics, Mathematical Modelling, Economics and Entrepreneurship (business decision-making, accounting statements, time value of money)

Experience

- **TU Wien – Christian Doppler Lab for Recommender Systems** — Vienna, Austria
  Software & DevOps Engineer (Student Employee), CDL RecSys — *Oct 2026 – Present*
  - **Conversational Recommender**: Develop and maintain an **LLM**-based conversational recommender system, a research prototype built with the lab's scientific staff and industry partners.
  - **Data & Integration**: Build API connections and data pipelines, and integrate and curate the data sources that ground the system.
  - **System Administration**: Administer the hosts, VMs, and services that the system runs on.
- **SRC d.o.o.** — Ljubljana, Slovenia
  Full Stack Developer — *Oct 2024 – Mar 2025*
  - **Delivery**: Led development of a **.NET / C#** microservices platform for internal and cross-enterprise document sharing and e-signing, with a Swagger/OpenAPI setup.
  - **Architecture & Requirements**: Ran requirements-gathering sessions directly with clients, then designed the system architecture and database end-to-end from those requirements.
  - **Mentoring**: Mentored an intern on frontend development.
- **SRC d.o.o.** — Ljubljana, Slovenia
  Software Engineer - Part time — *Jul 2023 – Oct 2024*
  - **Microservices Integration**: Built core features of a **.NET/Vue.js** visitor management microservice, integrating third-party APIs for QR code scanning and partner business data.
  - **Collaboration**: Held meetings and product demos with an external hardware partner, presenting progress and collaborating on integrating their hardware and software into the platform.
  - **Deployment**: Deployed the application on **Microsoft Azure**.
- **SRC d.o.o.** — Ljubljana, Slovenia
  Application Tester — *Aug 2022*
  - **QA**: Performed structured application testing using SoapUI, covering functional and API-level test cases.

Projects

- **MovRec – Conversational Movie Recommender**: Built a conversational recommender in **Python** with **FastAPI**, **PostgreSQL** and **pgvector**, grounded by the **Gemini API**. Gemini calls a semantic-search tool that runs a vector search over pre-computed movie embeddings (MovieLens, enriched with TMDb plot summaries); replies are grounded only in the retrieved movies. Runs via **Docker Compose**.
- **Combining 15 Stock-Return Forecasts (Erste Quant Challenge)**: Combined daily forecasts from 15 anonymised models for around 500 stocks into one long-only daily allocation in **Python** (**pandas**), with walk-forward validation (63-day refits, 11-day purge). Found two hidden signals: the difference of two near-identical models and an interaction where one model's skill flips sign with another. Out-of-sample IC rose from 0.039 to 0.087; validation Sharpe ratio 2.47 against 1.53 for the best single model.
- **Erste Group Earnings Volatility – Event Study**: Measured Erste Group share-price volatility around its 11 quarterly results dates since 2023 in **Python** (**pandas**, **yfinance**), benchmarked against the ATX. Volatility on the publication day is about 2.8 times its baseline, and none of 1,000 random placebo date sets came close; volatility is back to normal within about five trading days.
- **Deep Reinforcement Learning for Market Prediction (Bachelor's Thesis)**: Designed and compared feedforward, LSTM, and GRU-based Deep Q-Networks in **Python** to predict buy/hold/sell decisions directly from historical price data across multiple financial instruments (2019 – 2024); adapted and tuned the reward function and state representation.
- **Industrial AI – Process Logic Benchmarking (Zero One Hack 2026)**: Built and trained transformer models in **Python** for sequence prediction and anomaly detection on time-ordered process data; trained and scheduled via SLURM on the Leonardo EuroHPC cluster; evaluated with Top-k accuracy, MRR, F1, and ROC-AUC using **scikit-learn**.
- **Teapot – AI Quality Control for Fiber-Optic Trench Documentation (ViennaUp Hackathon 2026)**: Built a computer-vision pipeline in **Python** that screens construction-site images for compliance and quality-control anomalies; collaborated in a small team under hackathon time pressure and presented findings to judges.
- **Vowi 2.0 & Notepilot**: Built two Retrieval-Augmented Generation applications in **Python** using **ChromaDB**, one with the **Gemini API** as the underlying language model; Notepilot is a note-taking assistant.
- **Analytics & Reporting with R Shiny**: Built interactive dashboards in **R Shiny** during Master's coursework to visualize and report on analytical and statistical results.
- **Bachelor's Coursework Projects**: Built and served backend APIs with **FastAPI** across several coursework projects, connecting them to database and data-processing layers.

Skills

- **Programming**:
  - **Proficient:** Python, SQL, C#, R
  - **Familiar:** C, Java
- **LLM & RAG**: Retrieval-Augmented Generation, Gemini API (tool calling, embeddings), ChromaDB, pgvector
- **Quantitative & ML**: pandas, scikit-learn, TensorFlow, Reinforcement Learning (DQN, LSTM, GRU), Transformers, statistical modeling, time-series data, walk-forward validation, event studies, R Shiny
- **Data & Backend**: PostgreSQL, FastAPI, REST API integration, Swagger/OpenAPI
- **DevOps & Tools**: Docker, Docker Compose, Microsoft Azure, Jenkins, Git, GitLab, SLURM/HPC
- **AI-Assisted Tools & Continuous Learning**: Claude Code, Cursor, active Kaggle competitor

Languages

- **Slovenian**: Mother tongue **English**: C1 **German**: B1

Extracurricular Activities

- **Board of European Students of Technology (BEST) Vienna**: Full member since December 2025; logistics responsible for beWANTED 2026 career fair and Summer Course 2026.
- **Competitive Swimming**: National-level competitor in Slovenia (2008 – 2021), training 5–10 times per week; gold and bronze medals at international competitions.